GEOMETRY · QUADRILATERALS AND POLYGONS

# Polygon & Quadrilateral Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Polygon & Quadrilateral Explorer** on Mr. K's Classroom: Geometry → Quadrilaterals and Polygons → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

## 1 Predict before you touch the tool

1. How many triangles do the diagonals from one vertex of a hexagon make?

_____

_____

## 2 Explore use the tool and record what you see

**Polygon angle sums**

| n | Name | Triangles | Interior sum | Each interior (regular) | Each exterior (regular) |
|---|---|---|---|---|---|
| 3 | triangle | | | | |
| 4 | quadrilateral | | | | |
| 5 | pentagon | | | | |
| 6 | hexagon | | | | |
| 8 | octagon | | | | |
| 9 | nonagon | | | | |
| 12 | dodecagon | | | | |

**Quadrilateral detective**

Use the shape buttons, then check each property.

| Shape | Parallel pairs | All sides ≅ | 4 right ∠s | Diagonals bisect | Diagonals ≅ | Diagonals ⊥ |
|---|---|---|---|---|---|---|
| Parallelogram | | | | | | |
| Rectangle | | | | | | |
| Rhombus | | | | | | |
| Square | | | | | | |
| Kite | | | | | | |
| Isosceles trapezoid | | | | | | |

## 3 Write the rule fill in each blank from what you observed

1. The interior angles of an n-gon add up to _____.

2. The exterior angles of any convex polygon add up to _____.

3. The diagonals of every parallelogram _____ each other.

4. A rectangle's diagonals are _____. A rhombus's diagonals are _____.

5. A square is both a _____ and a _____.

## 4 Check no tool: show your work

1. Find the interior angle sum of a 10-gon.

2. Each exterior angle of a regular polygon is 24°. How many sides?

3. Find each interior angle of a regular 20-gon.

4. A quadrilateral's diagonals bisect each other and are perpendicular but not congruent. Name it.

## 5 Reflect

Why do the exterior angles always add to 360°?